

MYP 1 Integrated Sciences – Term 1 Revision Paper

Criterion A – Knowing and Understanding -15 Marks

Q1. Two students measure the length of the same object, whose accepted length is **20.0 cm**.

Student A records: 18.2 cm, 18.3 cm, 18.2 cm

Student B records: 19.8 cm, 20.2 cm, 20.0 cm

Outline which student's measurements are more **precise** and which are more **accurate**. Give a reason for each answer. **2 Marks**

Q2. Complete the statements using appropriate scientific terms.

4 Marks

- a. The SI unit of length is the _____.
- b. The amount of matter in an object is called _____.
- c. The SI unit of temperature is _____.
- d. Measurements that are close to the true or accepted value are described as _____.

Q3. A student measures the volume of water as **36.786 mL**.

2 Marks

Apply your scientific knowledge to round the measurement to **two decimal places** and **describe** why measurements should be rounded appropriately when recording experimental data.

Q4.A student observes:

Situation

Observation

A Candle wax melts when heated and becomes solid again when cooled.

B A piece of magnesium burns and forms a white powder.

Interpret the observations to identify the **physical change** and the **chemical change**. Use evidence to justify your answer. **3 Marks**

Q5. Three observations are recorded:

Substance	Initial state	Final state
Ice	Solid	Liquid
Water droplets on a cold glass	Gas in air	Liquid
Dry ice	Solid	Gas

A student states:

“Every substance must pass through the liquid state when changing from a solid to a gas.”

Interpret the information and make a **scientifically supported judgment** about the statement. Use evidence and appropriate changes-of-state terminology. **4 Marks**

Criterion B – Inquiring and Designing -14 marks

Q6. Investigating the Melting of Ice

A student notices that an ice cube melts more quickly in a warm room than in a cool room. The student wants to investigate **how temperature affects the time taken for an ice cube to melt**.

6.1 Outline a clear, testable **research question**.

2 marks

6.2 Outline a **testable prediction** and support it using scientific reasoning about particles.

2 marks

6.3 Outline the:

3 marks

- i. independent variable
- ii. dependent variable
- iii. two controlled variables

6.4 Design a logical and safe investigation to test the prediction. Include the **materials/apparatus with appropriate quantities**, a clear step-by-step method, how the data will be collected, and **two safety precautions**.

7 marks

Criterion C – Processing and Evaluating -11 Marks

Q7. Investigating Dissolving

A student investigates how **water temperature affects the time taken for 10 g of sugar to dissolve in 100 mL of water.**

Temperature (°C)	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Mean time (s)
20	124	120	122	
30	101	99	100	
40	78	81	78	
50	61	59	60	
60	43	41	42	

Hypothesis: *If the temperature of the water increases, the time taken for sugar to dissolve will decrease.*

7.1 Calculate the mean time for each temperature and **present** mean dissolving time against temperature on an appropriate graph. Include a title, labelled axes with units, appropriate scale and accurately plotted points.

7.2 Interpret the data and graph to describe the relationship between temperature and dissolving time. Use

at least two numerical values as evidence and explain the pattern using particle theory.

7.3 Discuss whether the hypothesis is supported. Use **two pieces of numerical evidence** and give a conclusion about the validity of the prediction.

7.4 Discuss one strength and one limitation of the method.

Criterion D – Reflecting on the Impacts of Science -10 Marks

Q8. Evaporation: Useful but Not Always Simple

Evaporation is the process in which a **liquid changes into a gas from its surface**. It happens when some liquid particles gain enough energy to escape into the air. Evaporation happens faster when the **temperature is higher**, the **surface area is larger**, or when there is more **air movement**.

People use evaporation in many ways in everyday life. Clothes dry because water evaporates from the wet fabric. Farmers also use evaporation to obtain **salt from seawater**. Seawater is placed in large, shallow ponds called salt pans. Heat from the Sun causes the water to evaporate, while the dissolved salt remains behind and can be collected.

Using solar energy for evaporation has environmental benefits because sunlight is a **renewable source of energy** and does not require burning fossil fuels. However, large salt pans require considerable land and may affect nearby habitats.

Evaporation also has social and economic importance. Salt production provides jobs and supplies salt for food and industry. However, workers may spend long hours outdoors in hot conditions, and salt production depends strongly on suitable weather. During rainy or humid conditions, evaporation becomes slower, reducing salt production.

Therefore, evaporation is a useful scientific process, but its benefits and limitations need to be considered.

8.1 Summarize how the science of evaporation is used in salt production. Include at least three relevant scientific points from the information provided **4 Marks**

8.2 Describe and summarize the positive and negative implications of using evaporation for salt production in relation to both of the following:

- **Environmental factor**
- **Social/economic factor**

Use relevant examples or evidence from the information provided to support your answer-6 Marks
